EXPARIMENT-1A

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main()
{
    int fd;
    char buffer[50];
    fd = open("sample.txt", O_CREAT | O_RDWR, 0644);

    if (fd < 0)
    {
        printf("File opening failed\n");
        return 1;
    }
    write(fd, "Hello OS Lab", 12);
    lseek(fd, 0, SEEK_SET);
    read(fd, buffer, 12);
    printf("Data read from file: %s\n", buffer);
    close(fd);
    return 0;
}
```

Lab1.c syscalls.c Syscalls1b.c

Assistant Server ~ Terminal File Edit View Go Help

```
~$ gcc syscalls.c -o system
~$ gcc syscalls.c -o syscalls
~$ ./syscalls
Data read from file: Hello OS Lab
~$ ./syscalls
```

EXPARIMENT-1B

```
#include <stdio.h>
#include <dirent.h>

int main()
{
    DIR *d;
    struct dirent *entry;
    d = opendir(".");
    if (d == NULL)
    {
        printf("Directory cannot be opened\n");
        return 1;
    }
    printf("Directory contents:\n");
    while ((entry = readdir(d)) != NULL)
    {
        printf("%s\n", entry->d_name);
```

```
    }
    closedir(d);
    return 0;
  }
```

```
bash: Syscalls1b.c: command not found
~$ gcc Syscalls1b.c -o syscalls1b
~$ ./syscalls1b
Directory contents:
.
..
2026-01-20-file-4.term
syscalls.c
Lab1.c
sample.txt
syscall.c
Syscalls1b.c
.smc
.snapshots
.ssh
syscalls1b
.cache
2026-01-20-file-3.term
.bashrc
2026-01-20-file-2.term
2026-01-20-file-1
.bash_profile
~$
```

EXPARIMNENT-1C

```
#include<stdio.h>
#include<unistd.h>
#include<sys/types.h>
void main()
```

```
{
printf("Before declaring FORK.");
printf("The value of PID is : ");
printf("%d",getpid());
printf("The value of PPID is : ");
printf("%d",getppid());

pid_t pid=fork();
if(pid==0)//Child Process
{
printf("After declaring FORK.");
printf("The value of PID is zero.");
printf("The value of PID is : ");
printf("%d",getpid());
printf("The value of PPID is : ");
printf("%d",getppid());
}
else if(pid>0)
{
printf("After declaring FORK.");
printf("The value of PID is greater than zero.");
printf("The value of PID is : ");
printf("%d",getpid());
printf("The value of PPID is : ");
```

```
printf("%d",getppid());
}
else
{
printf("After declaring FORK.");
printf("The value of PID is less than zero.");
printf("The value of PID is : ");
printf("%d",getpid());
printf("The value of PPID is : ");
printf("%d",getppid());
}
}
```